

Numerical-method verification: interpretation report (‘medium’ run)

Run: numerics_verification_test_med_run at run mode medium, 2026-09-10, 17 min.

Preset: celltype_pairs_sfaEI_Sc0p2sig0p1_noise0p025_dualStd_3cond_mu8p25

($n = 500$, 4000 state variables, piecewise activation, $\sigma_u = 0.025$, three adaptation regimes). Data: data/numerics_verification_test_med/numerics_verification/numerics_verification_data.mat. Figures: fig_numerics_solver/Fig_numerics_solver.png, fig_numerics_lya_method/Fig_numerics_lya_method.png.

This report supersedes figs/numerics_verification_test/interpretation_report_fast.md, which describes the methods in full (reshooting, the three references, the shared-Brownian-path construction, and the stimulus-edge trap). Only what changed at ‘medium’ is restated here.

What ‘medium’ changed relative to ‘fast’:

setting	fast	medium
noise-free reshoot window (stimulus-on third of T)	2.3-3.99 s of 6 s	4.3-7.99 s of 12 s
noisy reshoot window	1.3-1.99 s of 3 s	1.8-2.99 s of 4.5 s
LLE run / accumulation	10 s / last 5 s	20 s / last 10 s
reduced network for QR	$n = 30$, $[-8, 4]$ s	$n = 40$, $[-10, 10]$ s, 10 s accumulation
restarts per segment length (cap)	400	800

Two follow-up checks were run by hand after the stage and are reported in section 3; they are one-line `build_from_preset` calls, not part of the stage.

1. Results

1.1 Noise-free precision of SRA1 (Fig_numerics_solver, rows 2-3)

RMS error per state component over one 0.02 s Benettin segment, SRA1 reset to the ode45 (1e-10) reference at every restart:

regime	400 Hz	800 Hz	1600 Hz	slope	relative at 400 Hz (x)
no adaptation (chaotic)	3.5e-5	8.8e-6	2.2e-6	2.01	1.0e-5
single-timescale	2.5e-6	6.1e-7	1.5e-7	2.00	8.6e-6
multiple-timescale	8.7e-8	2.2e-8	5.4e-9	2.00	7.5e-7

Identical to ‘fast’ within 30%, with twice the restarts and a window that sits later in the run. Second-order convergence in every regime, in every state family.

1.2 With noise, shared Brownian path (row 3, dashed)

regime	400 Hz	800 Hz	1600 Hz	slope
no adaptation	4.8e-5	1.2e-5	3.0e-6	2.00
single-timescale	1.2e-5	3.0e-6	8.0e-7	1.95
multiple-timescale	4.6e-6	1.2e-6	3.4e-7	1.87

Same picture as ‘fast’: at $\sigma_u = 0.025$ the drift error dominates and the slope stays near 2. The mild droop toward 1.5 in the adapted regimes is the stochastic term becoming visible where the drift error is smallest.

1.3 LLE, ode45 vs SRA1, full network, 10 s accumulation (row 1)

regime	Benettin, ode45 1e-10	Benettin, SRA1 400 Hz	fast run
no adaptation	+4.015	+3.853	+3.645 / +3.715
single-timescale	+0.317	+0.578	+0.319 / +0.489
multiple-timescale	-0.1193	-0.1193	-0.212 / -0.201

Chaotic regime: 4% apart. Stable regime: identical to four digits. The single-timescale regime is again apart, now by 0.26; section 3 resolves it.

1.4 Benettin vs QR, reduced network, n = 40 (Fig_numerics_lya_method)

regime	states	Benettin LLE	QR lambda_1	difference
no adaptation	40	+0.0576	+0.0558	0.002
single-timescale	160	-0.4007	-0.4010	0.0003
multiple-timescale	320	-0.1143	-0.1209	0.007

All three agree. The two points that were open at ‘fast’ are closed:

- The 40-neuron no-adaptation network is no longer a dead fixed point (the 30-neuron one had LLE $-9.9 = -1/\tau_d$). At $n = 40$ it sits at the edge of chaos with an exponent near zero and a local exponent that swings between -15 and $+15$ (row 1, left), which is the demanding case for both methods, and they agree to 0.002.
- The single-timescale gap at ‘fast’ (-1.10 vs -0.50 , over 4 s on a transient) has vanished with 10 s of accumulation: -0.4007 vs -0.4010 . The local-exponent panel shows why the short window was unreliable: the trajectory is a periodic burst train whose local exponent spikes to $+5$ every ~ 0.7 s and dips to -6 between bursts, so a 4 s average is dominated by how many bursts happen to fall inside it.

The QR spectra (row 2) are plausible in every case: a single near-zero leading exponent, a plateau near $-1/\tau_d = -10$ for the 40-state network, and staircases at the adaptation and depression rates for the others.

2. Interpretation

The integrator. Everything the ‘fast’ run said is confirmed with twice the data and a later window. At the paper’s 400 Hz, SRA1 makes a relative error of $1e-5$ (chaotic), $9e-6$ (single-timescale) and $8e-7$ (multiple-timescale) per state over one Benettin segment, converging at second order with or without the paper’s noise. In the contracting regimes this saturates at a few $1e-5$ per component and stays there; in the chaotic regime no integrator can track a trajectory for more than a few Lyapunov times, and the statistic that survives, the LLE, agrees between SRA1 and a $1e-10$ adaptive reference to 4% there and to four digits in the stable regime.

Benettin vs QR on SRNNCellTypePairs is now verified. With adequate accumulation the two independent methods agree to better than 0.01 in all three regimes, on a network that spans a near-zero exponent, a periodic burst train and a stable state. The class’s Benettin implementation and its QR implementation (which uses the analytic Jacobian) therefore corroborate each other, as they already did on `SRNNModel2`.

3. The single-timescale LLE gap between integrators is finite-time variability, not bias

The one number that did not settle at ‘medium’ was the full-network single-timescale LLE: ode45 $+0.317$ vs SRA1 $+0.578$. Two hand checks discriminate between “SRA1 is biased” and “these are two different trajectories with different finite-time exponents”.

Refining the step does not move SRA1 toward ode45. Same seed, same window, noise off:

integrator	LLE
SRA1 400 Hz	+0.578
SRA1 800 Hz	+0.711

integrator	LLE
SRA1 1600 Hz	+0.691
ode45 tol 1e-10	+0.317
ode45 tol 1e-6	+0.328

A step-size bias would shrink monotonically with the step; it does not. Note also that the three ode45 values (0.319 at ‘fast’, 0.317, 0.328 at loose tolerance) are one trajectory: their mutual error is small enough that they have not separated within 20 s at this exponent. Each SRA1 run, whose per-segment error is 1e-6 rather than 1e-10, is a distinct trajectory on the same attractor by the time the window opens.

Across network seeds the spread is far larger than the gap, and its sign flips. SRA1 at 400 Hz vs ode45 at 1e-10, 20 s runs, last 10 s accumulated:

seeds	SRA1	ode45
[1 2]	+0.578	+0.317
[2 3]	+0.988	+0.827
[3 4]	+0.331	+0.610
[4 5]	+0.543	+0.430

Perturbing only the initial condition (same seed, `x0_std` 0.1001 instead of 0.1) gives +0.565, so nearby trajectories give nearby values and the sensitivity is to which trajectory, not to numerical noise.

The finite-time LLE of this regime over 10 s therefore varies by about 0.3 to 1.0 from realisation to realisation, and the ode45/SRA1 difference on any one seed lies well inside that spread, with either integrator coming out higher depending on the seed. This is the signature of an intermittent regime (the local exponent in row 1 of the LLE figure alternates between bursts of growth and quiet decay). It is a property of the dynamics, and it is exactly what the paper’s reps sweeps and their medians are for; it is not a defect of the integrator.

4. Verdict

- **SRA1 at 400 Hz: verified** (second order, 1e-5 to 1e-7 relative error per Benettin segment, LLE integrator-independent wherever the finite-time exponent is well defined).
- **Benettin vs QR on SRNNCellTypePairs: verified** (agreement to < 0.01 in all three regimes on the 40-neuron network at 10 s accumulation).
- **A caution for the manuscript, not a concern about the numerics:** in the single-timescale regime the 10 s finite-time LLE varies by a factor of three across seeds and trajectories. Single-realisation LLE values in that regime should not be quoted to better than about ±0.3; medians over reps should.

5. What to use in the manuscript

The ‘medium’ figures in this folder are the ones to cite; the ‘fast’ set is a plumbing check with shorter windows. If a supplementary statement is wanted, section 2 gives it in two sentences, and section 3’s two tables justify reporting medians over reps for the single-timescale regime.

6. Reproduction

```
setup_paths();
numerics_verification_test_med_run           % ~17 min at 'medium'
```

The follow-up checks in section 3 are `build_from_preset(preset, 'sfa1_std1', 'sigma_u_noise', 0, 'ode_solver', <'sra1'|'ode45'>, 'fs', <fs>, 'T_range', [0 20], 'lya_method', 'benettin', 'lya_T_interval', [10 20], 'rng_seeds', <seeds>)` followed by `run()`; for the tolerance variant set `ode_opts` before `run()`.

7. Follow-up: is the single-timescale gap an initial-condition effect, a `lya_dt` effect, or a window effect?

Three controls, all Benettin on the full 500-neuron network, seed [1 2], noise off, 400 Hz, both integrators. Script: `scripts/examples/lle_finite_time_variability.m` (about 15 min).

7.1 Same network, same stimulus, different initial conditions

`x0_std` scaled by $1 + k/100$, which rescales the same initial draw and leaves W and the stimulus untouched. 20 s runs, last 10 s accumulated:

IC scale	SRA1 400 Hz	ode45 1e-10
1.00	0.578	0.317
1.01	0.627	0.592
1.02	0.714	0.633
1.03	0.332	0.680
1.04	0.630	0.591
mean	0.576	0.563

A 1% change of the initial condition moves either integrator's 10 s exponent by up to 0.35. The two five-member ensembles overlap completely and their means agree to 0.01. The 0.317 that started the question is the low tail of ode45's own distribution, not a property of ode45.

7.2 Renormalisation interval

<code>lya_dt</code> (s)	SRA1	ode45
0.01	0.5779	0.3172
0.02	0.5779	0.3174
0.05	0.5780	0.3175
0.10	0.5780	0.3172

No effect to four digits for either integrator. This also disposes of the hypothesis that the finite Benettin perturbation crossing the piecewise activation's breakpoints biases the estimate: a five-fold longer interval changes how often that happens and changes nothing.

7.3 Longer accumulation

window	SRA1	ode45
[10, 20] s	0.578	0.317
[10, 40] s	0.566	0.697
[10, 60] s	0.567	0.659

SRA1 is already stable at 30 s. ode45 leaves its 10 s value behind and settles near 0.66-0.70. The gap has reversed sign and shrunk from 0.26 to 0.09. If the finite-time scatter shrinks as $1/\sqrt{T}$ from about 0.2 at 10 s, the expected scatter at 50 s is about 0.09, so the remaining gap is about one standard deviation.

7.4 Interpretation

The single-timescale regime is intermittent: its local exponent alternates between bursts of growth and quiet stretches of decay (see row 1 of the LLE figure and the burst train on the reduced network). A finite-time Lyapunov exponent on such a regime depends on which stretch of the attractor the trajectory visits during the window. Two runs that differ by anything at all, $1e-6$ of integrator error or 1% of initial condition, are independent samples of that

distribution once a few Lyapunov times have passed. The data say the distribution has a standard deviation near 0.15-0.2 over 10 s and a mean near 0.57 on this network, and that both integrators sample the same distribution.

Consequences for the manuscript:

- A single-realisation LLE in the single-timescale regime should not be quoted to better than about ± 0.2 over a 10 s window. The reps sweeps and their medians are the right estimator, and the paper’s practice of reporting medians across reps is what makes the regime comparisons meaningful.
- The chaotic and stable regimes do not have this problem. Their local exponents are steady, so a single 10 s window gives an exponent that is integrator-independent to 4% and to four digits respectively.
- Nothing here changes the conclusion about the integrator: SRA1 at 400 Hz is second-order accurate with per-segment relative error $1e-5$ to $1e-7$, and its ensemble-mean LLE matches a $1e-10$ reference to 0.01.

7.5 Remaining concerns

1. **Ensemble means agree to 0.01 on five initial conditions; that is a small sample.** The claim “same asymptotic LLE for both integrators” is supported by 7.1 and by the reversal in 7.3, but a two-sided test would want 20-30 initial conditions. The sweeps’ reps already provide this for SRA1; the ode45 half would cost about a minute per realisation.
2. **The 50 s ode45 value (0.66) sits 0.09 above SRA1 (0.57).** This is consistent with one standard deviation of finite-time scatter, but it was not shown to shrink further, and a longer window with ode45 at $1e-10$ costs about 3 s per simulated second on this network. If a reviewer asks whether SRA1 underestimates the exponent in this regime, the answer should come from 7.1-style ensembles, not from one long run.
3. **Intermittency itself.** The 10 s finite-time exponent varying by a factor of two on one network is a physical property of the single-timescale regime worth a sentence in the manuscript, since it bears on how “edge of chaos” should be read there. It has not been characterised beyond these five samples.
4. **The reduced network used for the QR comparison is not the paper’s network.** At $n = 40$ it reproduces the three regimes qualitatively (edge of chaos, periodic bursting, stable) and both Lyapunov methods agree on it, but that is a check of the methods, not of the 500-neuron exponents. QR on the full network would need a different algorithm (a partial spectrum) and is not planned.
5. **Noise regime of SRA1.** The strong-order-1.5 term never became visible at $\sigma_u = 0.025$ because the drift error dominates. That is the reassuring direction, but it means the noisy sub-experiment verifies the drift, not the stochastic increment handling, on this network. The stochastic handling is verified separately at large σ in `test_sde_integrators` (measured order 1.73).